

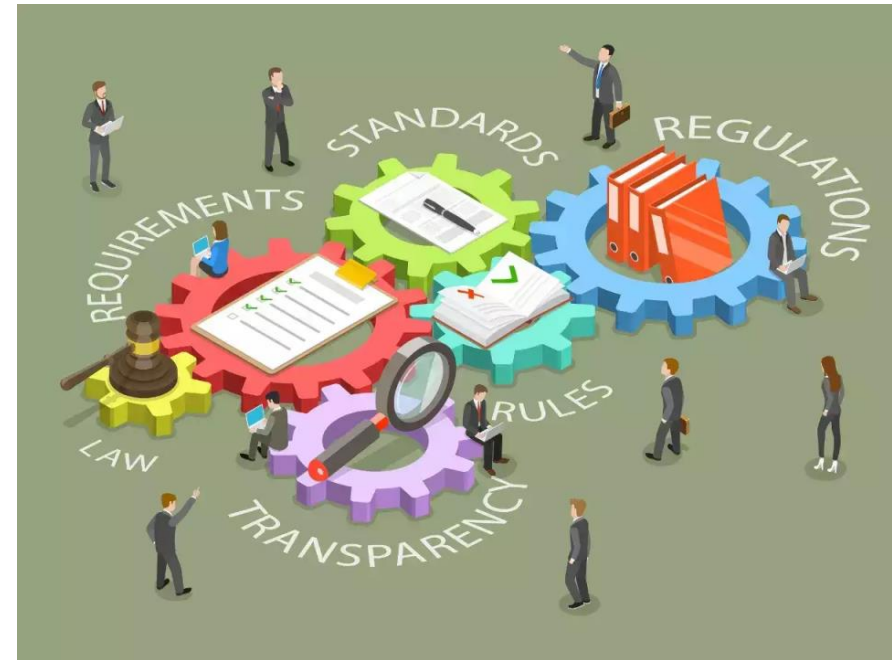
Microeconomics II



Topic 6: Industry equilibrium

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Motivation



Roadmap

1. Market equilibrium under perfect competition
2. Comparative statics, applications

Market demand curve

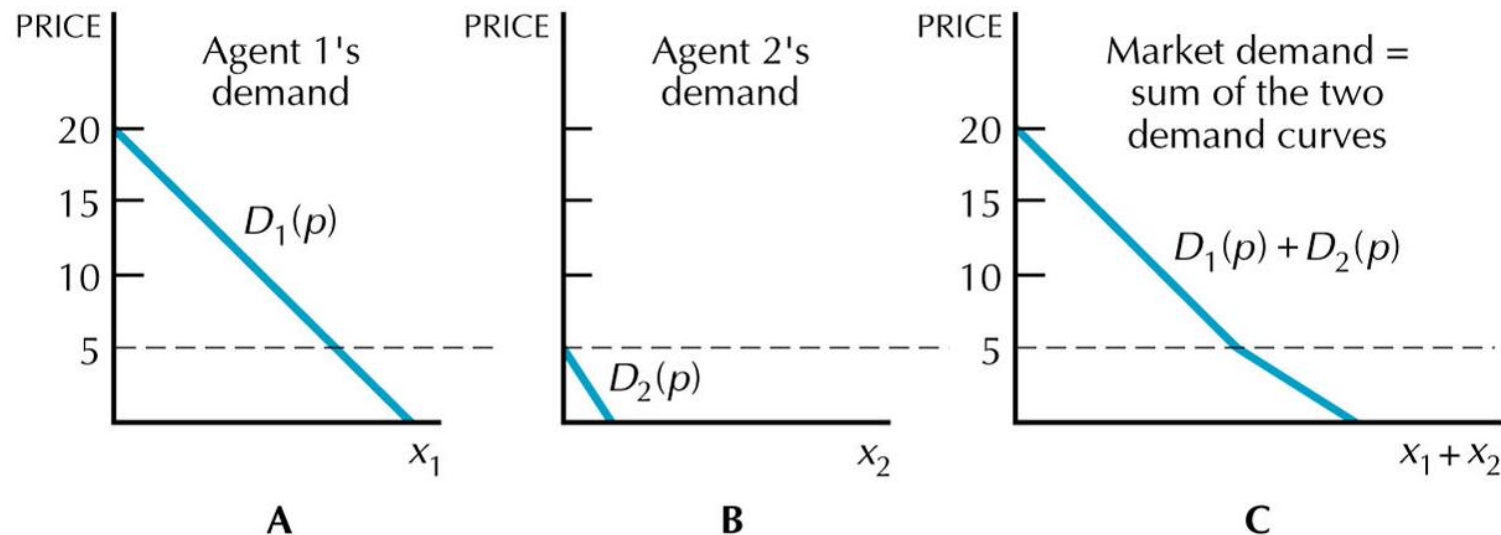
Aggregate demand: sum of all individual demands

- ▶ $X^1(p_1, p_2, m_1, \dots, m_n) = \sum x_i^1(p_1, p_2, m_i)$
- ▶ Depends on prices and distribution of incomes.
- ▶ For representative consumer: $X^1(p_1, p_2, M)$

Fix p_2 and income: construct market demand curve.

- ▶ Higher p_2 : shift outward for substitutes, inward for complements.
- ▶ Higher income: shift outward for normal good.

Inverse demand function: horizontal sum of individual functions.



Market demand curve

The slope determined by:

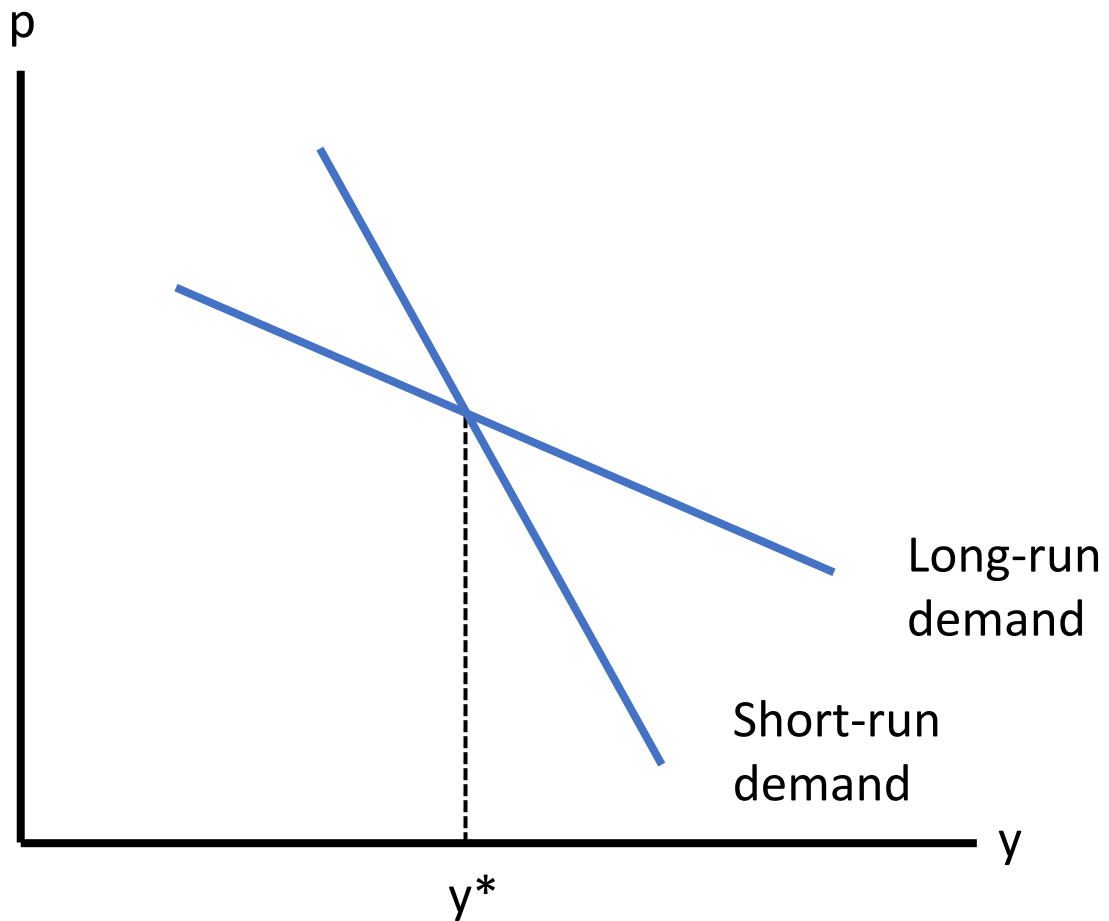
- ▶ Adjustment on the intensive margin.
 - ▶ Negative for ordinary good.
 - ▶ May be positive if, at some price levels, the good is Giffen good for large number of consumers.
- ▶ Adjustment on the extensive margin (number of consumers).
 - ▶ Marginal consumer

Price goes down: people buy more and the number of consumers increases.

Usually, the market demand is downward sloping.

Long-run vs. short-run market demand curve

→ Demand curve more elastic in the long run than in the short run



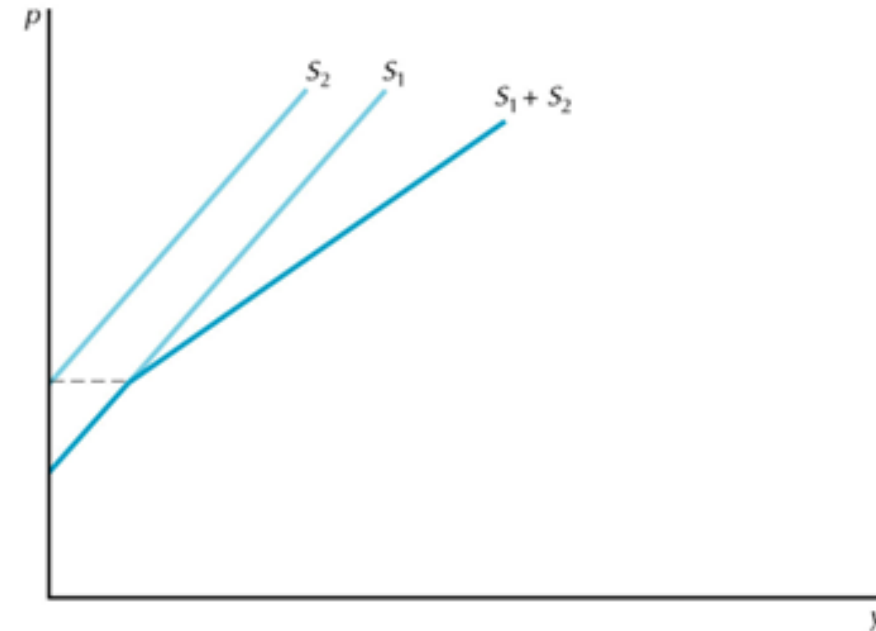
Short-run market supply curve

Important distinction between short- and long-run.

- ▶ Nature of inputs and decision-making process.
- ▶ Entry to and exit from the industry.

Short-run industry supply curve: sum of individual supply curves.

- ▶ Fixed number of firms.
- ▶ $S(p) = \sum S_i(p)$
- ▶ Graphically, horizontal sum of inverse supply curves.



Short-run market supply curve



Exercise

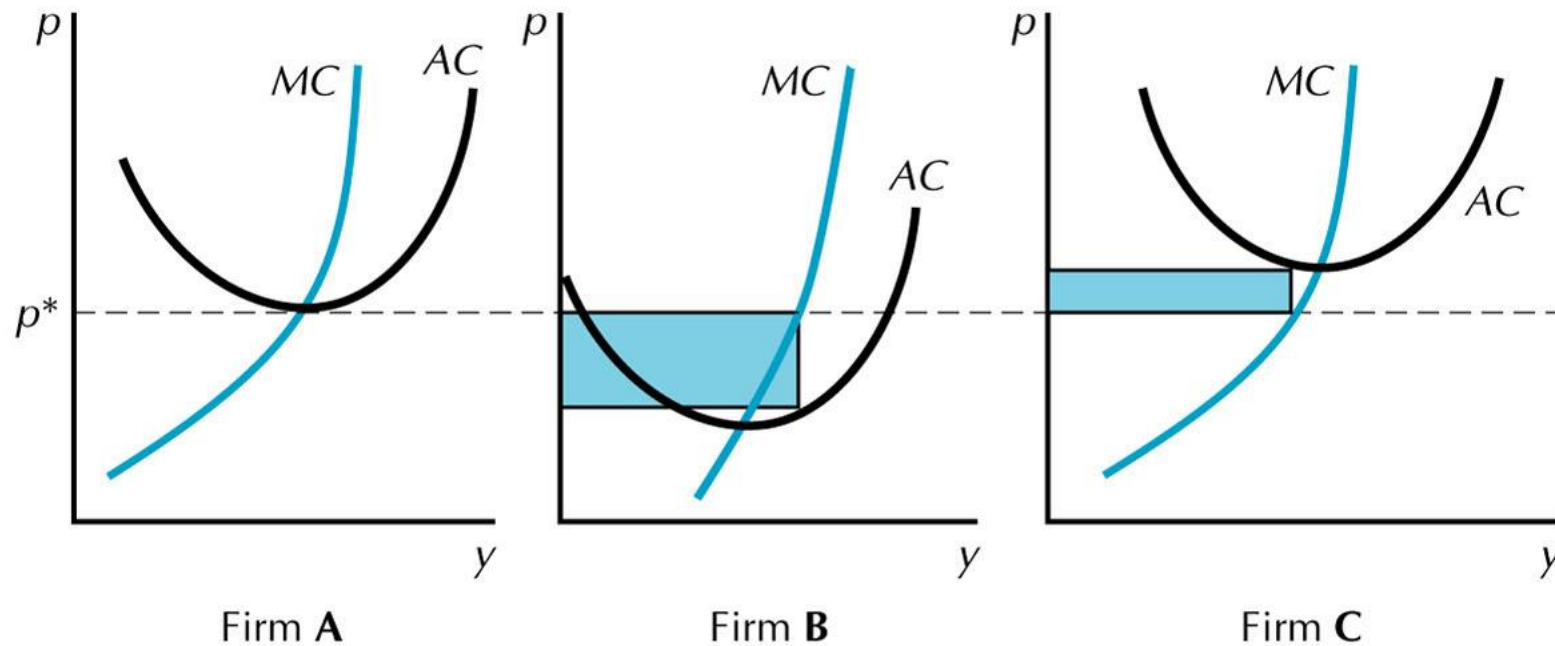
In the short-run, there are 100 firms in the industry with identical cost functions: $STC(y) = y^2 + 30y + 400$

- ▶ Derive the industry supply $S(p)$.
- ▶ What is the output produced by the industry at $p = 80$?

Short-run equilibrium

Intersection of the market supply and market demand curves.

Some firms make zero profit, some positive, some negative.



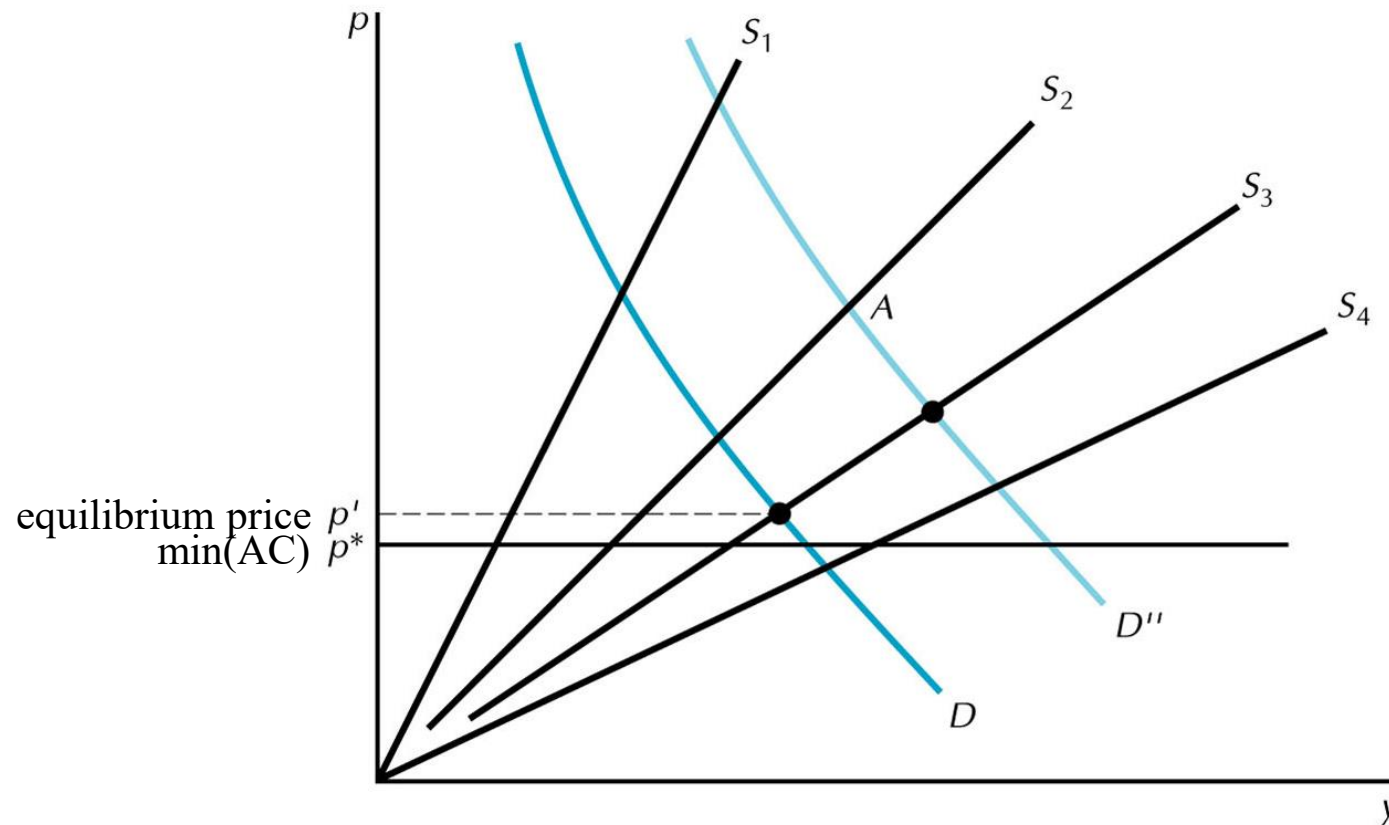
Long-run equilibrium

Free-entry industry

- Most competitive industries have free entry (but some feature entry barriers)
 - Negative profits → exit
 - Positive profits → entry
- shifts in market supply curve and equilibrium market price
- In long-run equilibrium, further entry unprofitable

Long-run equilibrium

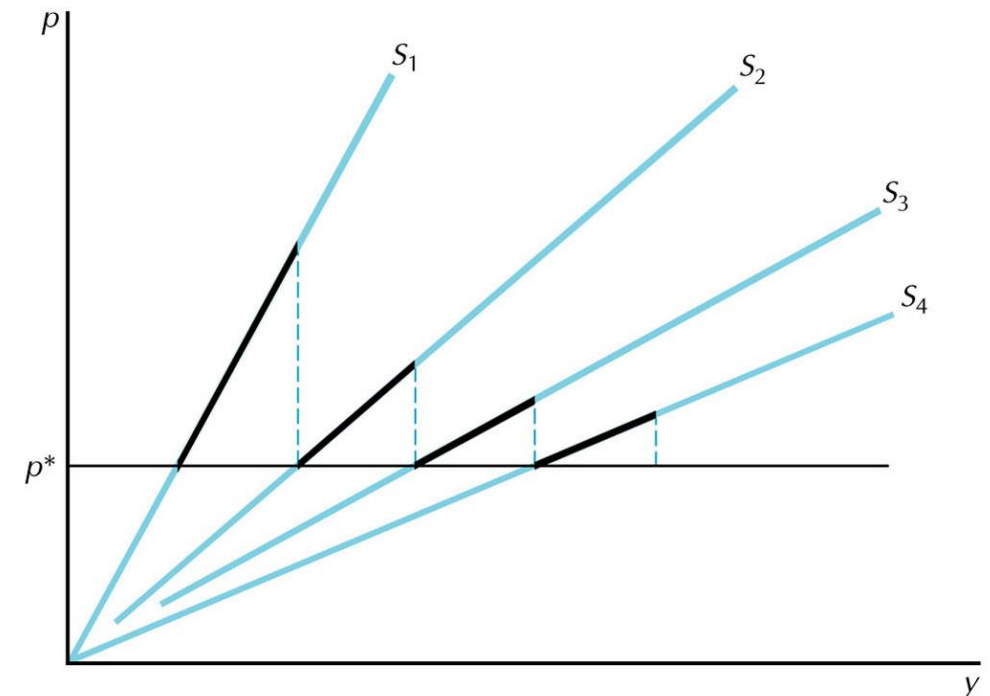
- The figure shows supply curves for different number of firms in the industry
- If a 4th firm enters, profits are negative → the equilibrium number of firms is 3



Long-run market supply curve

≠ sum of individual long-run supply curves because of free entry and exit

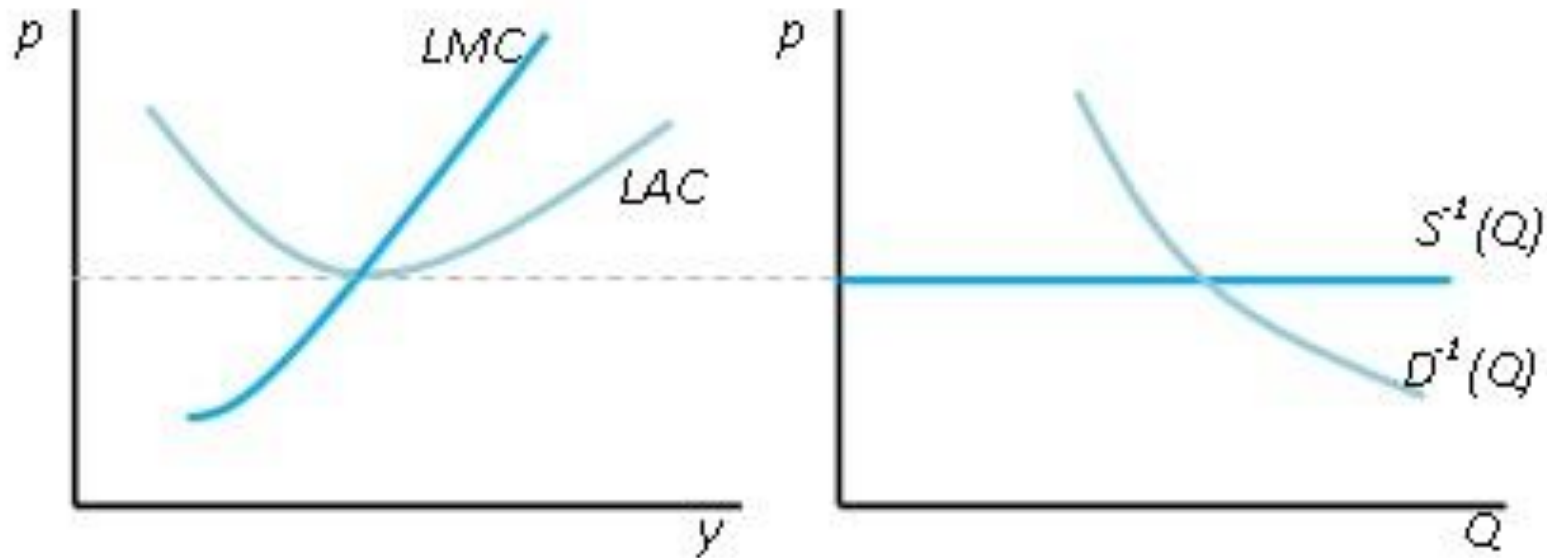
- Market supply curve is not at
 - Points below market price p^*
 - Points like A in the previous figure
 - Demand going through the point would intersect supply with additional exit above p^*
- Black segments show candidate points
- With more firms in the industry, the segments get flatter
- With many firms, given by minimum ACs of operating firms
- If the firms are identical,
 - Long-run market supply curve horizontal
 - Zero profits
 - Output determined by market demand
 - Market output adjusted exclusively via entry/exit



Long-run market supply curve

Concave-convex LTC curve, U-shaped LAC curve

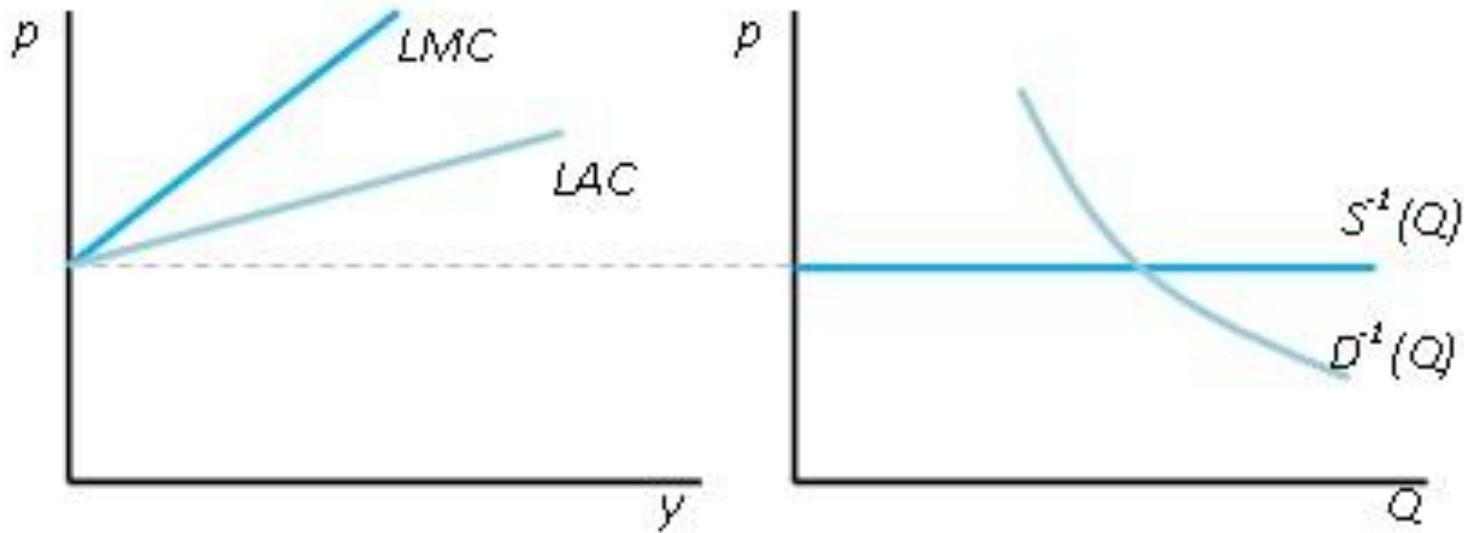
- Firms willing to produce positive output and earn zero profit



Long-run market supply curve

Convex LTC curve

- LMC and LAC increasing at all output levels
- Infinite number of firms, each producing infinitely small output



Long-run market supply curve

Concave LTC curve

- LMC and LAC decreasing at all output levels

→ monopoly

Long-run market supply curve: Exercise



Exercise

Firms with identical cost function: $LTC(y) = 2y^3 - 30y^2 + 150y$.

Market demand $D_q(p) = 1500 - 4p$.

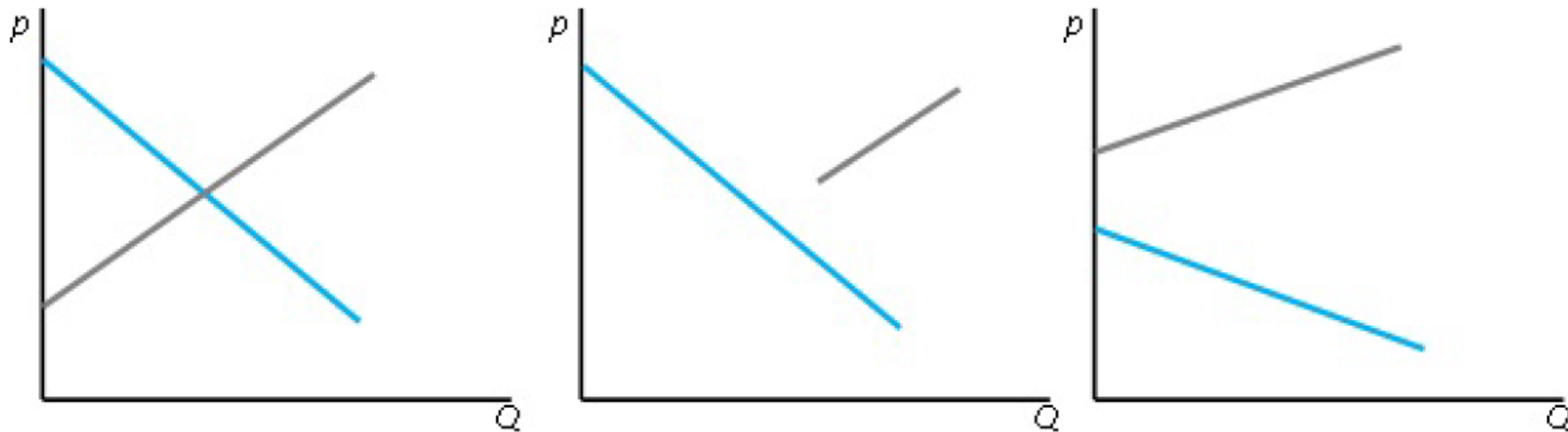
- ▶ Determine the inverted long-run supply curve $p = S_Q^{-1}(y^*)$.
- ▶ Determine the total output supplied by the industry in the long-run.
- ▶ What is the equilibrium number of firms in the industry?

Equilibrium

Interaction of market demand and supply: equilibrium market price.

- ▶ Graphically: point where the curves cross.
- ▶ Algebraically: $D(p^*) = S(p^*)$ or $P_S(q^*) = P_D(q^*)$

Existence of equilibrium



Roadmap

1. Market equilibrium under perfect competition
2. Comparative statics, applications

Entry and exit: Application

Hsieh and Moretti, 2003. Can Free Entry be Inefficient? Fixed Commissions and Social Waste in the Real Estate Industry, *Journal of Political Economy*.

Residential real estate brokerage in US.

- ▶ Commissions the same everywhere. It seems better to be an agent in cities with high house prices.
- ▶ However, average salary is the same everywhere.
- ▶ Reason: free entry and exit.
 - ▶ An agent in a city with high house prices sells fewer houses.
 - ▶ In such a city, more people decide to become real estate agents.
 - ▶ The study found this pattern across cities, as well as within cities over time.

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**Vaši nemovitost
(byt, RD, chatu, pozemek)
prodáme nejrychleji,
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Víte o někom, kdo chce koupit
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nám to a my Vám za to zaplatíme!

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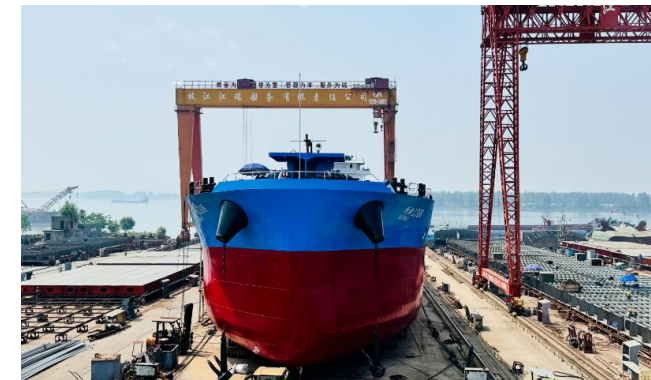
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External pecuniary economies of scale

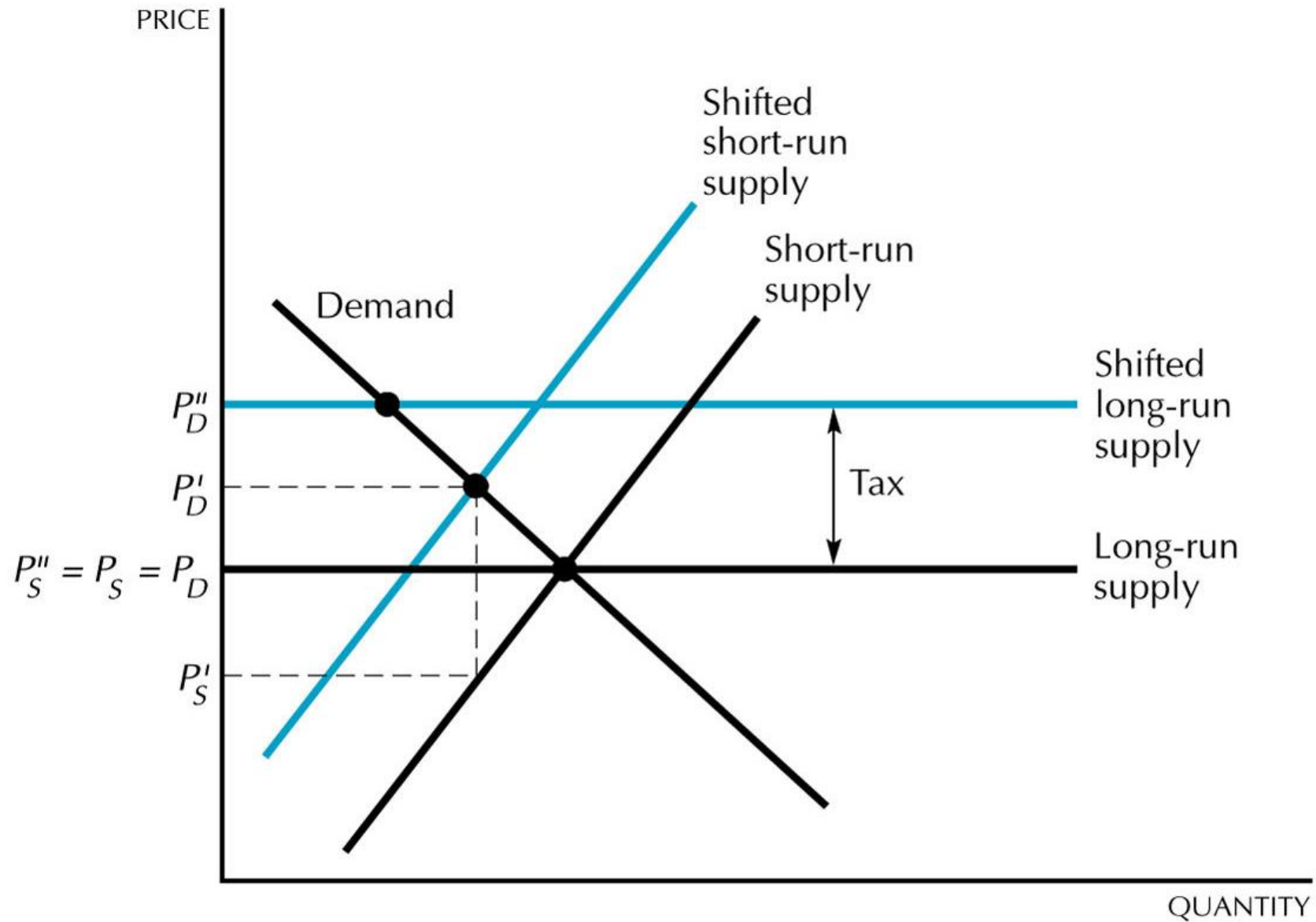
Costs not constant when the output of the industry goes up (due to change in the demand of the industry for inputs).

Individual firms cannot affect the input price, whole industry can.

- ▶ Negative external pecuniary economies of scale: input becomes more expensive.
 - ▶ Example: metal as an input.
- ▶ Positive external pecuniary economies of scale: input becomes cheaper.
 - ▶ Example: stronger bargaining power of more firms against suppliers who have increasing returns to scale.



Taxes



Economic rents

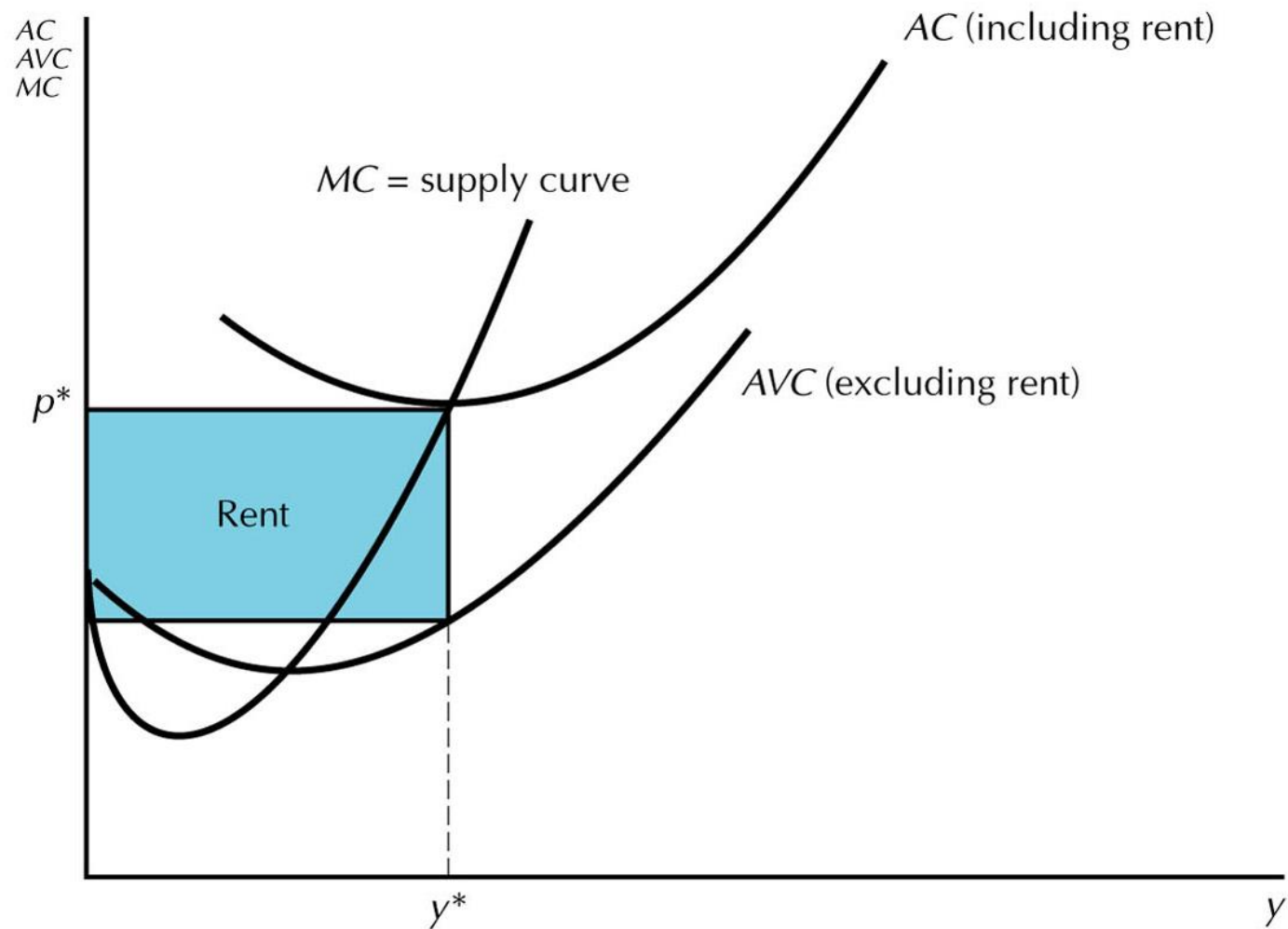
- Some factors of production are fixed
 - Fixed by nature: e.g. land, natural resources, talent...
 - Fixed administratively: e.g. licences, taxis, liquor stores
- This gives rise to economic rents

Economic rents = payments to a factor of production that is in excess of the minimum payment necessary to have the factor supplied.

- This can lead to positive financial profits but still no entry
- The reason is that entry is based on economic profits that include the market rental rate for the fixed factor of production
- **Rent-seeking** = efforts directed at keeping or acquiring claims to factors in fixed supplies
 - E.g. lobbying expenses, lawyers' fees, public relations costs
 - Socially pure deadweight loss

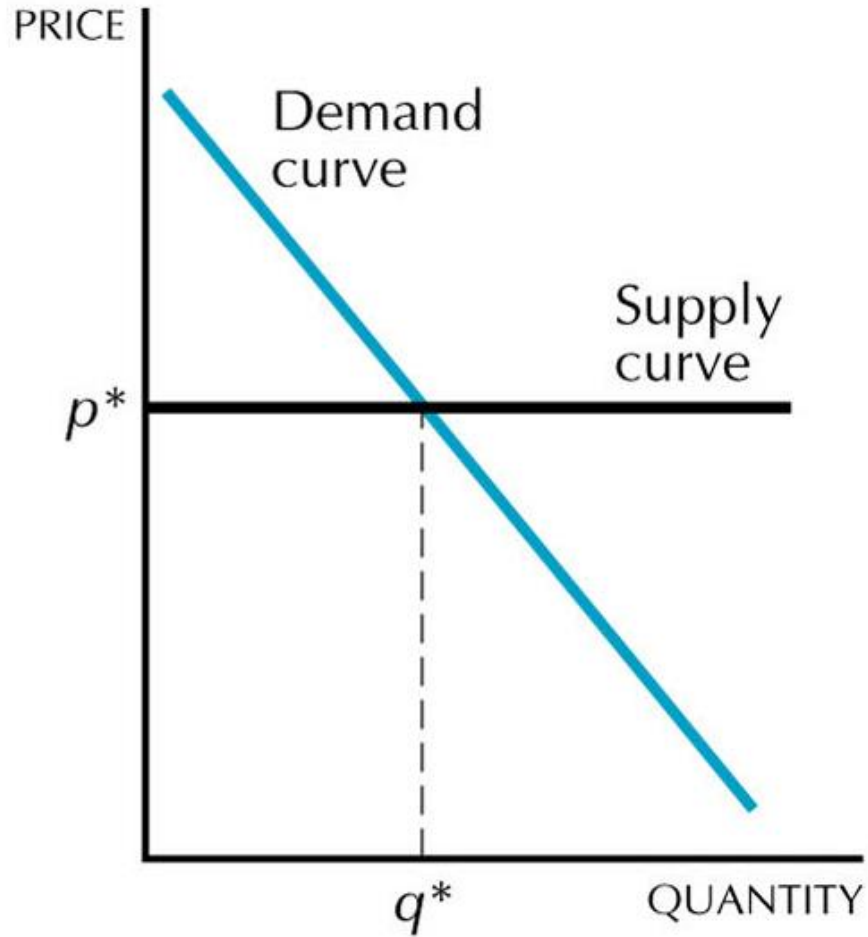


Economic rents



Economic rents = producer surplus

Equilibrium with horizontal supply curve

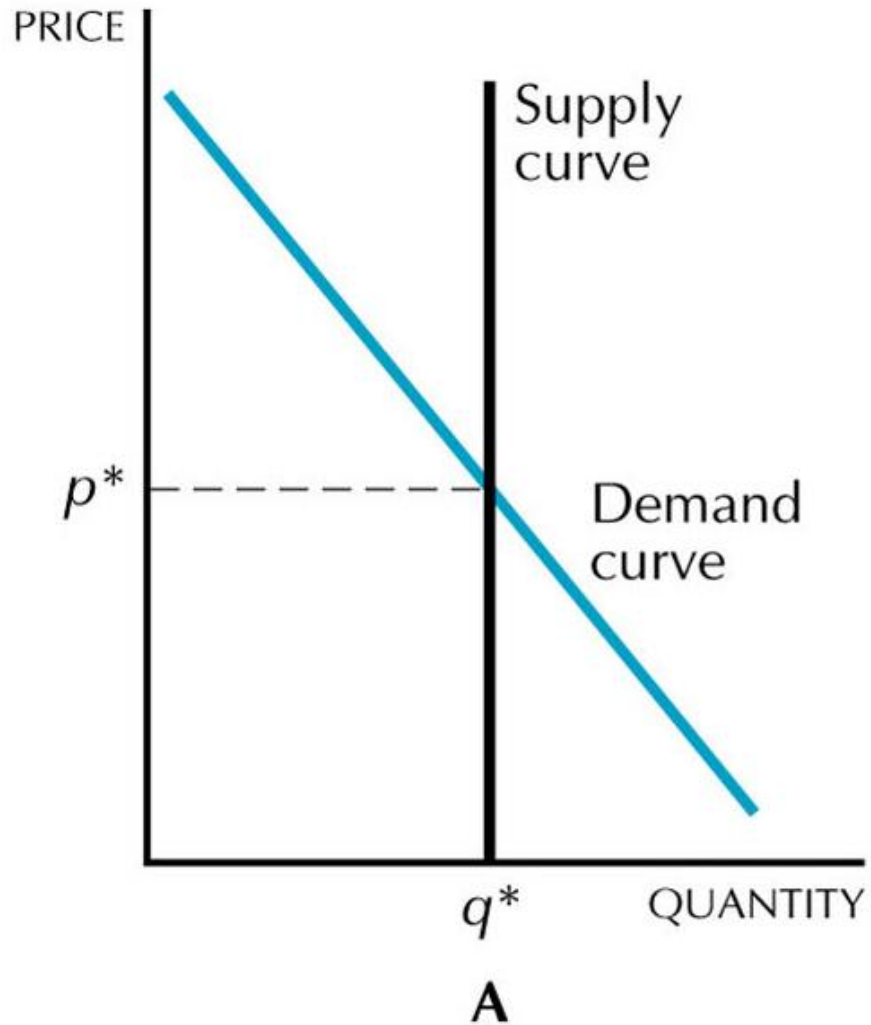


B

- Price determined by supply
- Quantity determined by demand



Equilibrium with vertical supply curve

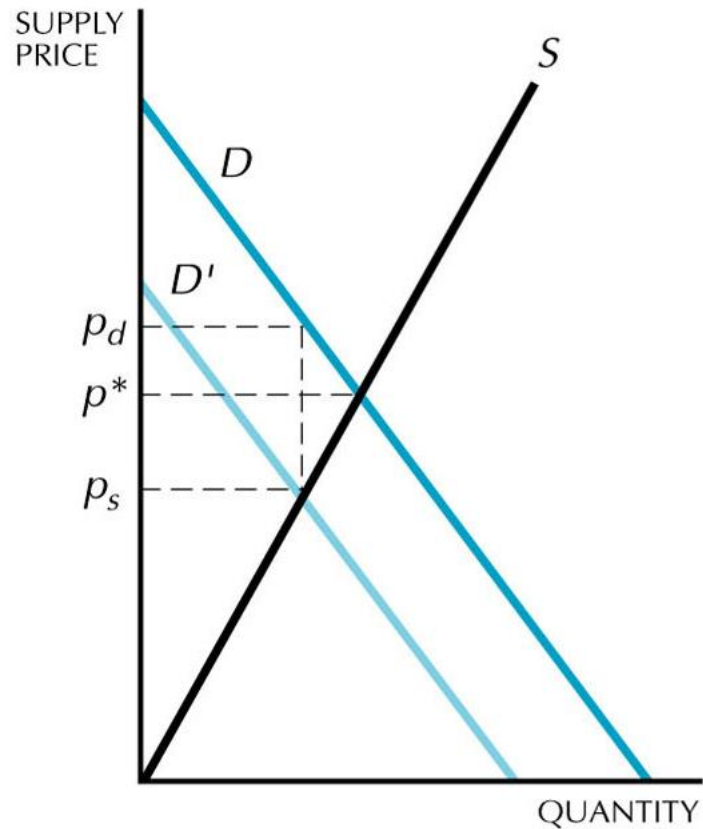


- Quantity determined by supply
- Price determined by demand

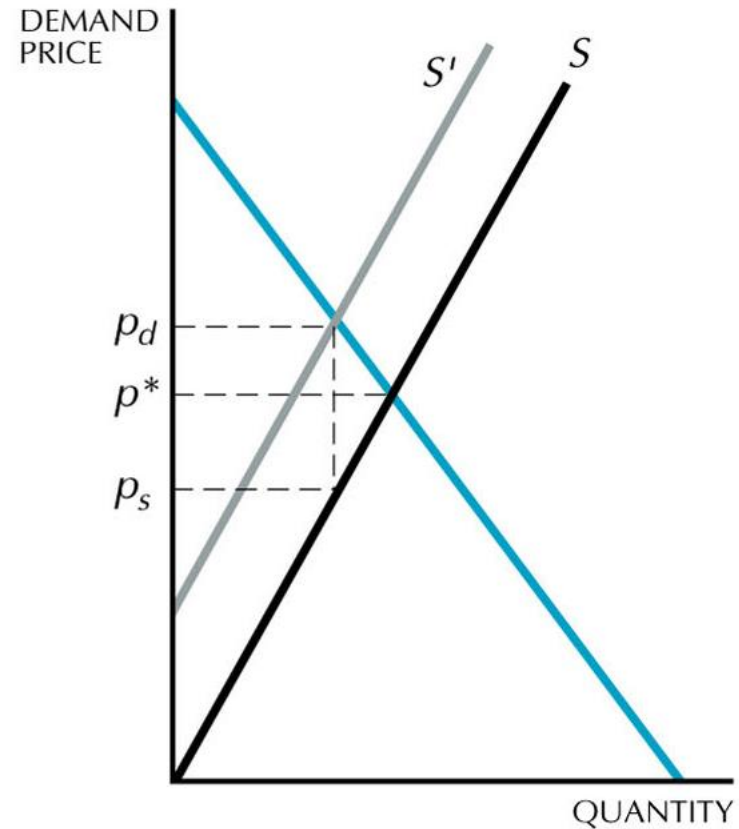


Equilibrium – comparative statics

Does it matter who pays the tax?



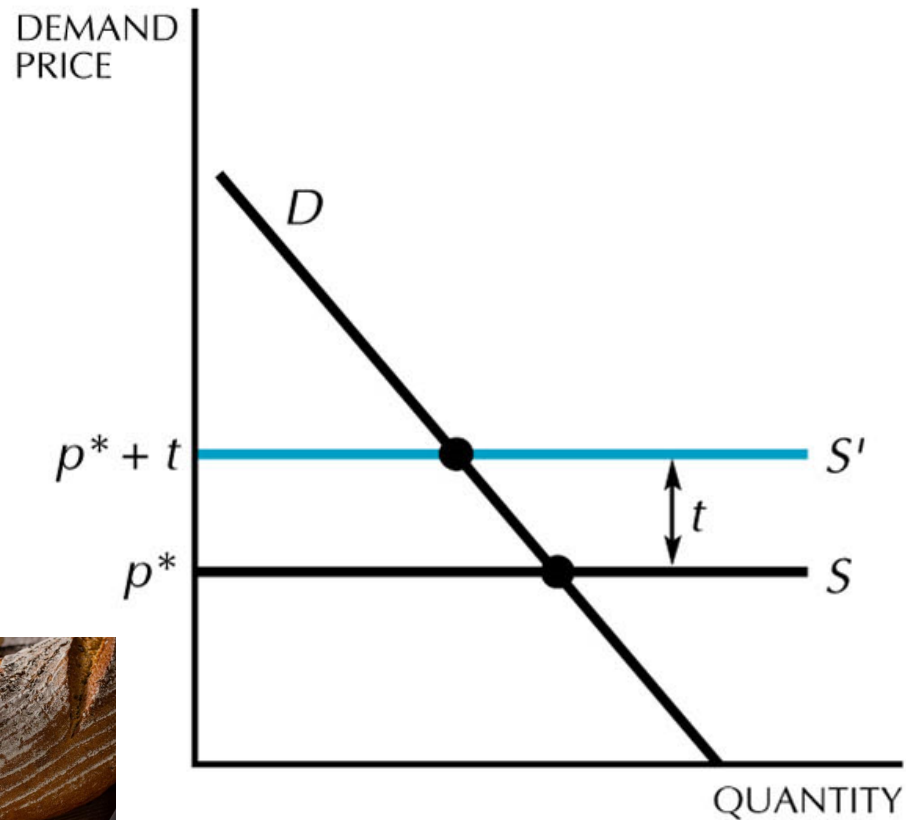
A



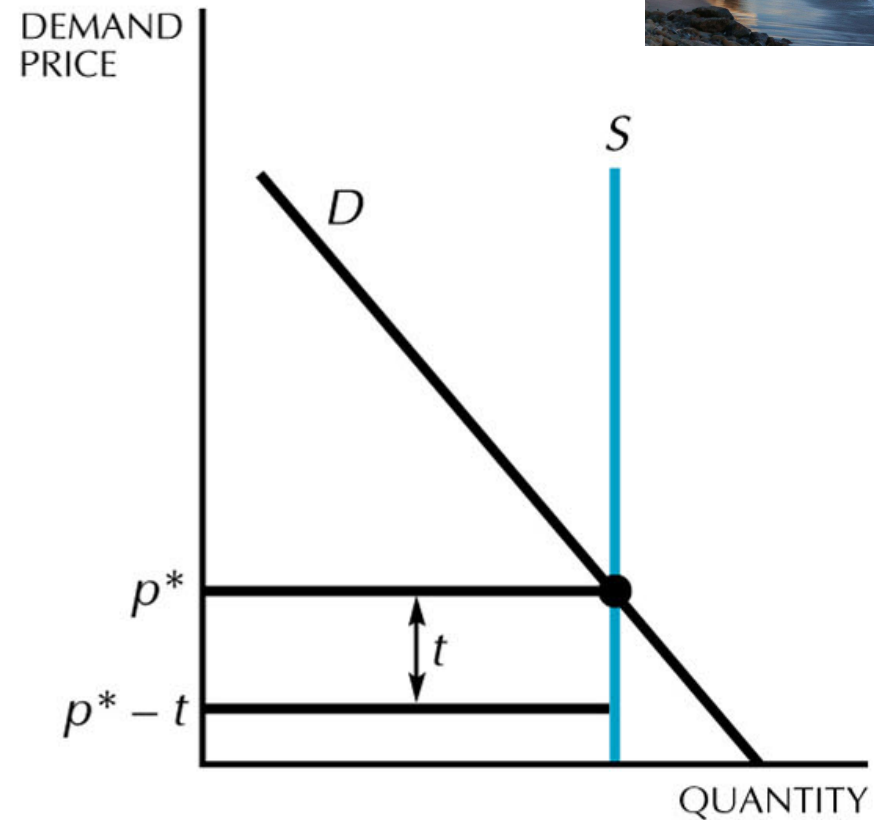
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Equilibrium – comparative statics

Who actually bears the cost of a tax?



A

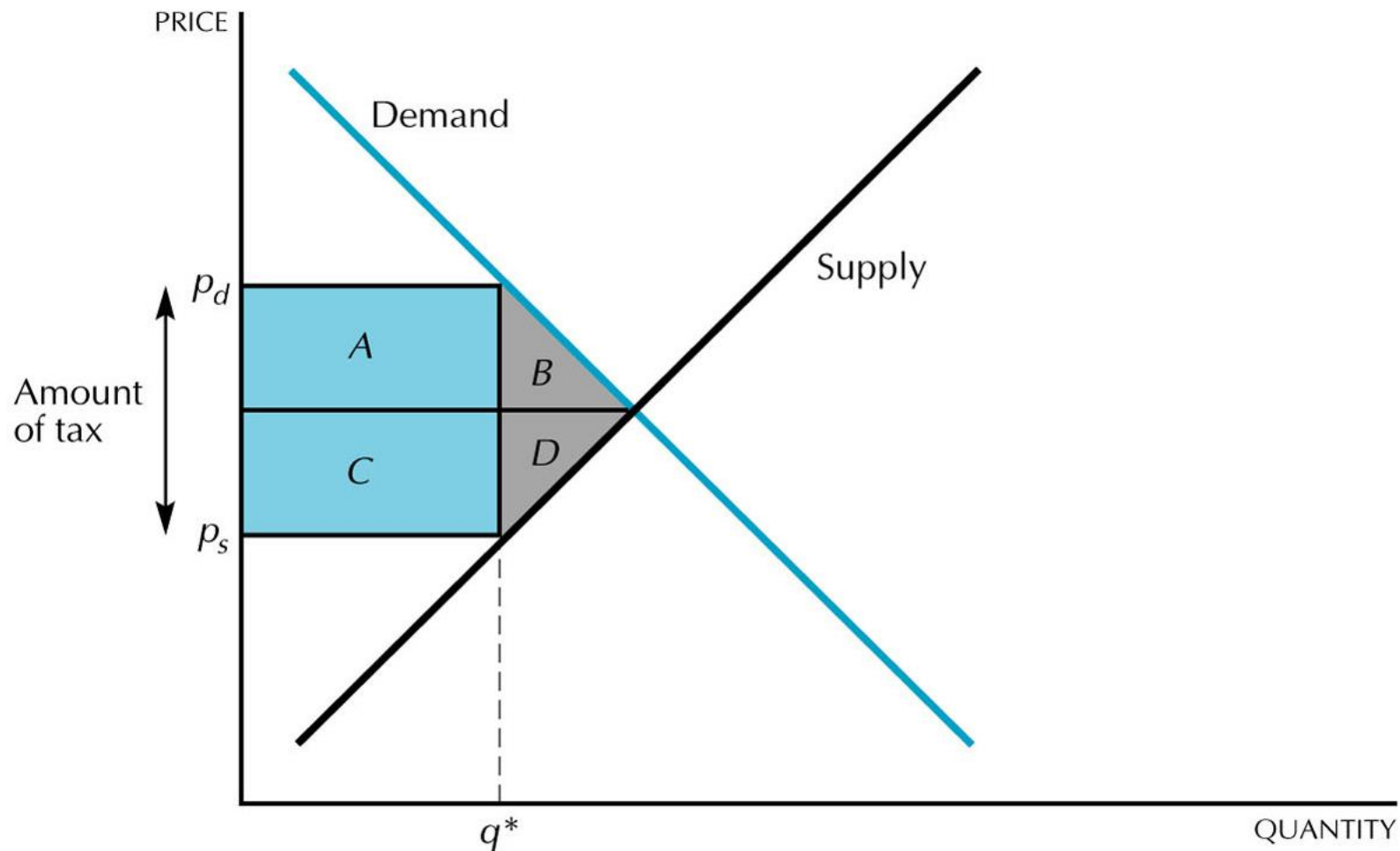


B



Equilibrium – comparative statics

Deadweight loss (excess burden of tax)



Equilibrium – comparative statics

Ways to cut down on prostitution

- ▶ Arrest prostitutes
- ▶ Embarrass their clients

www.chicagopolice.org/pr/

Nejnavištbovanýjst Jak začít Přehled zpráv Ubytování.net

Haga clic aquí para la versión en español.

The Chicago Police Department in conjunction with the Mayor's office have now made prostitution solicitors' information available online. By using this website, you will be able to view public records on individuals who have been arrested for soliciting prostitutes or other related arrests.

The following individuals were arrested and charged for either patronizing or soliciting for prostitution. It is not a comprehensive list of all individuals arrested by the Chicago Police Department for patronizing or soliciting for prostitution. The names, identities and citations appear here as they were provided to police officers in the field at the time of arrests.

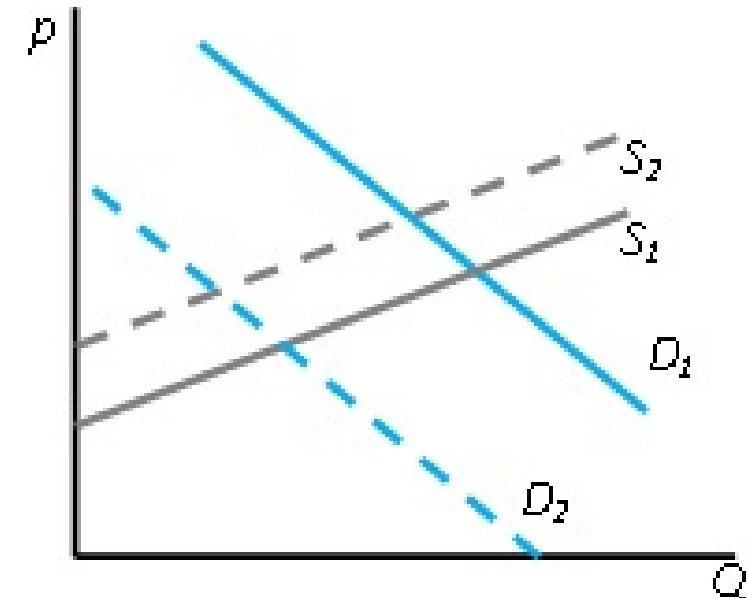
Click [here](#) to see a list of the statutes and their descriptions.

All photos and information will automatically be removed after thirty (30) days from the arrest date. These individuals are presumed innocent until proven guilty in a court of law.

District of Arrest ALL Change Timespan Last 5 Days Submit Change

SEARCH RESULTS KEY

NAME	HAYES, WILBURT
SEX/AGE	M/51
HOME ADDRESS	1401 COMMONS DRIVE
HOME CITY	WOODSTOCK
ARREST ADDRESS	158 E 5TH ST
ARREST DATE (Y/M/D)	2013/06/02
STATUTE	720 ILCS 5.0/11-15-A-1
VEHICLE IMPOUNDED(Y/N)	N



Equilibrium – comparative statics

Minimum wage

- ▶ Higher wage
- ▶ Excess supply of workers, unemployment. Critics argue this is partially responsible for incidence of crime among teenagers.
 - ▶ Let the wage back to market level.
 - ▶ Subsidies to firms who hire teenagers.

Exercise

- ▶ The mayor of a city thinks that if people get jobs, vandalism will reduce. At the same time thinks the market wage is too low.
- ▶ She keeps the minimum wage and pays subsidy for each worker hired ($w^{min} - w^a$), where w^{min} is the minimum wage and w^a is the market clearing wage.
- ▶ Each unemployed causes damage $w^v - w^a$ where $w^v > w^{min}$.
- ▶ Will the subsidy save the city money?

Perfect competition and efficiency

Pareto efficiency: any economic agent cannot get better off unless another one gets worse off.

- ▶ Price equals marginal cost and marginal willingness to pay. Nobody willing to pay for additional unit the amount for which somebody would be willing to produce it.
- ▶ Ratio of marginal utility and price is the same for all consumers. They cannot get better off by exchange of goods.
- ▶ Ratio of marginal product and price of inputs are the same for all producers. They cannot get better off by substituting one input for another.
- ▶ $p \cdot MP_i = w_i$ for all inputs. Producers cannot get better off by using more or less of an input.
- ▶ $p = LAC_i$ in each industry. Producers cannot get better off by moving into another industry.

Unrelated to distribution of welfare and inequality.

Group discussion

Please get in groups of 3.

1. You are advising a government that hopes to reduce drug use. How do the policies you would recommend depend on the elasticities of supply and demand for drugs?
2. Economists often recommend taxing factors that give rise to economic rents. Propose 2 different reasons why this might be the case.

